Lesson plan · for the teacher

Fitting a line

Data science · Level 3 · Lesson 1

Time
40 min

Materials
Board and chalk or markers; Paper and pencils; A phone or tablet with Darsi, for the lab

Learning goals

- Fit a line to a scatter plot.
- Read the slope and the start.
- Know where a line stops working.

Activity: A ruler line

1. Warm-up (5 min): Ask a tall pupil and a short pupil to stand up. Ask: whose arms are longer, and how could we predict anyone's?
2. Teach (8 min): Draw a scatter of reading hours and marks, lay a line and read off its slope and start.
3. Plot (15 min): Pupils measure in pairs and add dots; volunteers lay a ruler where the line fits best.
4. Predict (7 min): Use the line to predict a teacher's arm span from their height, then measure it.
5. Wrap-up (5 min): Ask two pupils to say the finding in one sentence with a number in it. Then give the exit ticket.

Discussion

- What does the slope mean here?
- Where would our line stop working?
- Was the prediction close?

Exit ticket

- mark = 5 + 3 × hours. Predict the mark for 2 hours.

Try it at school tomorrow
Plot height against arm span on the board and let pupils lay a ruler where the line fits best. Predict one new pupil, then measure and compare.

Practice questions · for you

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1 **Our pupils read 0 to 5 hours. Should we trust the line for 40 hours a week?**

◯ Only on Mondays ◯ No, it is far outside the data ◯ Yes, lines go on forever

2 **Use the line mark = 8 + 2 × hours.**

A pupil who reads 4 hours is predicted _____ marks. The slope is _____ marks per hour.

Word bank: 8 · 16 · 2 · 12

3 **Work out the slope and the start from the data, then print the prediction for 6 hours.**

```
hours = [1, 2, 3, 4, 5]
marks = [10, 12, 14, 16, 18]
mx = sum(hours) / 5
my = sum(marks) / 5
slope = 0
start = my - slope * mx
print(round(start + slope * 6, 1))
```

Write your program:

4 **A line of best fit proves that reading more causes higher marks.**

◯ True ◯ False

5 **Our data covers 0 to 5 hours of reading. Which prediction can we trust most?**

◯ 20 hours ◯ 3 hours ◯ 40 hours

✎ **Which prediction at your school could a simple line make? What would its limits be?**

Check your answers

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1 **No, it is far outside the data**

A line only knows the range of data it learned from.

2 **A pupil who reads 4 hours is predicted 16 marks. The slope is 2 marks per hour.**

8 + 2 × 4 = 16, and the number multiplying hours is the slope.

3

```
hours = [1, 2, 3, 4, 5]
marks = [10, 12, 14, 16, 18]
mx = sum(hours) / 5
my = sum(marks) / 5
sxy = 0
sxx = 0
for i in range(5):
    sxy = sxy + (hours[i] - mx) * (marks[i] - my)
    sxx = sxx + (hours[i] - mx) ** 2
slope = sxy / sxx
start = my - slope * mx
print(round(start + slope * 6, 1))
```

The slope is 20 ÷ 10 = 2 and the start is 14 − 2 × 3 = 8, so 6 hours predicts 8 + 12 = 20.

4 **False**

A line describes a pattern. Other things, like help at home, could explain both.

5 **3 hours**

Three hours is inside the data; the others are far outside it.